INCH-POUND
MIL-PRF-29504/3C
w/AMENDMENT 1
30 April 2024
SUPERSEDING
MIL-PRF-29504/3C
19 January 2021

PERFORMANCE SPECIFICATION SHEET

TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENT RESISTING, DUMMY TERMINUS (FOR USE WITH MIL-PRF-28876 AND MIL-DTL-83526 CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-29504.

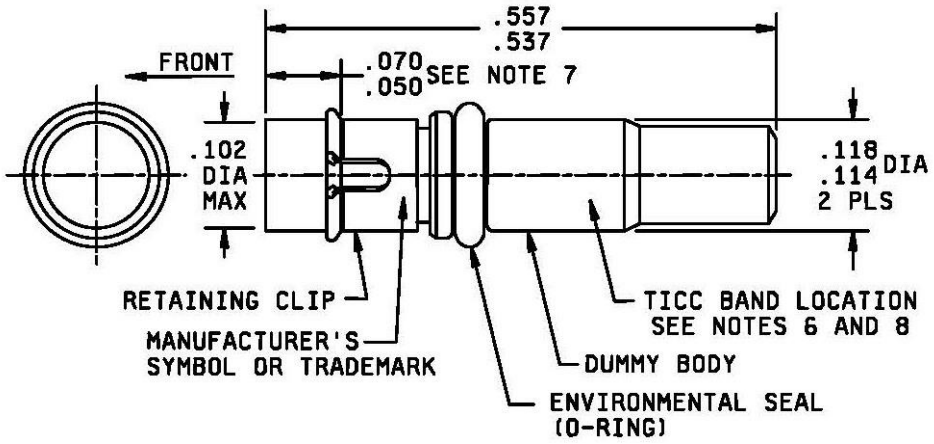
Part or identifying number (PIN): See table II below and 1.2 of MIL-PRF-29504.

M	**29504**	**/03**	**-**	**XXXX**
Prefix for Military Specification	Specification Number	Spec Sheet No.		Termini Identification Color Coding (TICC) Code (see table II)

AMSC N/A FSC 6060

Distribution Statement A. Approved for public release: distribution is unlimited.

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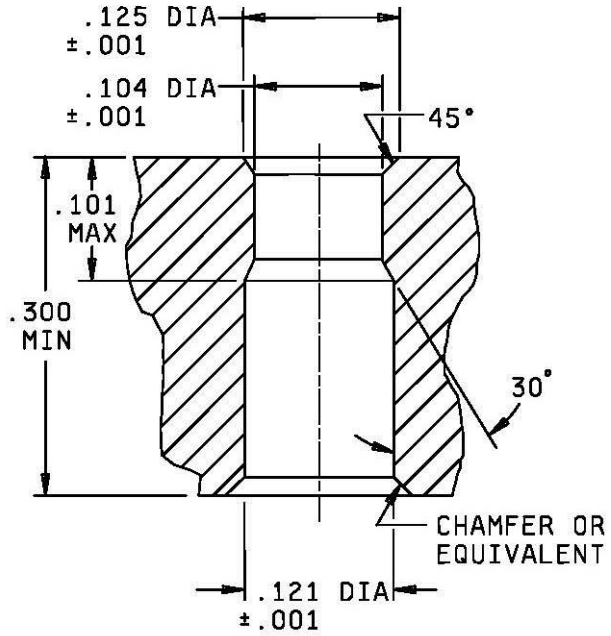
Inches	Millimeters
0.050	1.27
0.070	1.78
0.102	2.59
0.114	2.90
0.118	3.00
0.537	13.64
0.557	14.15

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters are to be concentric within 0.002 inch (0.05 millimeter).
4. Dimensions apply to plated/finished part.
5. Dummy termini to be used to fill unused cavities.
6. The TICC bands shall be at the rear of the dummy terminus.
7. Dimensions to be measured with terminus installed in connector insert or equivalent gauge fixture.
8. The dimension shown is effective prior to the application of the TICC marking.

FIGURE 1. Dummy terminus.

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Inches	Millimeters
0.001	0.03
0.101	2.57
0.104	2.64
0.121	3.07
0.125	3.18
0.300	7.62

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within 0.002 inch (0.05 millimeter).
4. Dimensions apply to plated/finished part.
5. Tolerance on all angles is ±1 degree, unless otherwise noted.

FIGURE 2. Insert equivalent fixture for terminus measurement.

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REQUIREMENTS:

Design and construction:

Interoperability: Applicable except insertion loss is not performed.

Dimensions and configuration: See figures 1 and 2 and table II.

Weight: 0.04 ounce (1.0 gram) maximum.

Adhesives: Not applicable.

Tools: See table I.

Insertion loss: Not applicable.

Optical discontinuity: Not applicable.

Circular runout: Not applicable.

Return loss: Not applicable.

Change in optical transmittance: Not applicable.

Terminus engagement and separation force: Not applicable.

Fiber pull-out: Not applicable.

Cable pull-out: Not applicable.

Usage: Termini compliant with this specification sheet may be used in connectors other than MIL-PRF-28876 at the discretion of the acquiring activity.

TABLE I. Tools.

Tool	Part number
Insertion tool	Drawing 499-6872813-2 (NSN 5120-01-144-5338)
Removal tool	Drawing 499-6872813-6 (NSN 5120-01-419-2942)

TABLE II. PIN and supersession data.

TICC	PIN	Superseded PIN	Superseded manufacturer's PIN	CAGE
-4038	M29504/03-4038	M28876/18-16	1143274S	53669

NOTE:

1. Allowance is granted for stock to be marked as "M29504/3-4038" for stock that is packaged prior to the dating of this revision to MIL-PRF-29504/3.
2. Qualified manufacturers have 30 days after publication of this revision to relabel existing QPL inventory packaging due to PIN change (example: from "M29504/3-4038" to "M29504/03-4038"). Relabeling of package marking for PIN doesn't require repeating group A conformance inspection. No other marking or date codes shall be changed.

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VERIFICATION:

Environmental/mechanical: Termini shall be tested to the following MIL-PRF-28876 environmental and mechanical requirements. Change in optical transmittance and optical discontinuity requirements are not applicable.

Impact

Vibration

Shock

Thermal shock

Temperature/humidity cycling

Temperature cycling

Temperature life

Flammability

Ozone exposure

Connector: The connector for this terminus shall be a qualified MIL-PRF-28876 connector.

Qualification by similarity: Manufacturers who are qualified under MIL-PRF-29504/14 or /15, and whose terminus passes size, weight, identification marking, and workmanship specified herein, are qualified under this specification sheet. This qualification by similarity is valid only if the same retention clip, same O-ring, same dimensions from the retention clip to the back end of the MIL-PRF-29504/14 and /15 terminus, and the same materials are used in the construction of this terminus as are used in the construction of the qualified /14 or /15 terminus.

Conformance inspection. Applicable for group A only, is valid only if the same retention clip, same O-ring, same dimensions from the retention clip to the back end of the MIL-PRF-29504/14 and /15 terminus, and the same materials are used in the construction of this terminus as are used in the construction of the qualified /14 or /15 terminus.

Group A. Group A shall consist of the inspections/tests specified for group A in MIL-PRF-29504.

Group A sampling plan. Group A inspection shall be performed by random sampling for products supplied under this specification. This random sampling shall be performed in accordance with MIL-STD-1678-1, requirement 1304. The number of samples for random sampling inspected per lot size is specified in the “general inspection” table of MIL-STD-1678-1, requirement 1304. The required acceptance criterion is a zero nonconformance regardless of sample size.

Group A sampling plan for occurrence of a nonconformance. If any nonconformance is found, then 100 percent inspection shall be used until the qualifying activity is satisfied that the cause for the nonconformance has been corrected and will not reoccur. Also, until the nonconformance has been corrected, inspection with the requirement that data be provided (in lieu of inspection by attribute) may be specified by the qualifying activity.

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
Navy − SH
(Project 6060-2023-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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